

Exhibit CC-123



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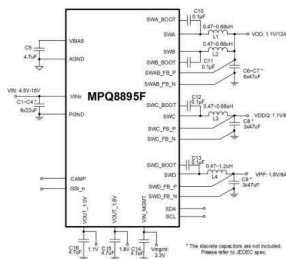
Products > Power Management > Power Management IC (PMIC) > MPQ8895F

MPQ8895F

12V, Power Management IC With I²C/I³C Interface for DDR5

Pre-Release

Datasheet



Description

The MPQ8895F is a complete power management IC (PMIC) that integrates four high-efficiency, step-down DC/DC converters and a flexible logic interface for DDR5.

TheMPQ8895F provides four channel buck outputs. Buck A and Buck B can be configured as a dual-phase interleaving buck converter with current-sharing performance. Buck C and Buck D can be configured as two single-phase buck converters that are designed to support the specific DDR5 power requirements.

The DC/DC converters provide constant-on-time (COT) control for fast transient response. The configurable 500kHz to 1250kHz switching frequency (f_{SW}) greatly reduces external inductor and capacitor size.

Full protection features include under-voltage lockout (UVLO), over-current protection (OCP), over-voltage protection (OVP), under-voltage protection (UVP), and thermal shutdown.

The MPQ8895F requires a minimal number of external components, and it is available in a space-saving QFN-35 (5mmx5mm) package.

Show less ^

Features & Benefits

- ✓ Wide 4.25V to 15V Input Voltage (V_{IN}) Range
- ✓ JEDEC PMIC50x0 Power Management IC (PMIC) Specification Rev 1.8 Compliant
- ✓ Adaptive Constant-On-Time (COT) for Ultra-Fast Transient Response
- ✓ Four High-Efficiency Step-Down Regulators:
 - Buck A: 6A DC/DC Converter
 - Buck B: 6A DC/DC Converter
 - Buck C: 6A DC/DC Converter
 - Buck D: 6A DC/DC Converter
- ✓ Buck A/B Supports Up to 12A of Continuous Current in Dual-Phase Mode
- ✓ I²C/I³C-Configurable Output Voltage (V_{OUT}) with Slew Rate Control:
 - SWA/B/C: 0.8V to 1.435V in 5mV Steps, or 0.6V to 1.235V in 5mV Steps
 - SWD: 1.5V to 2.135V in 5mV Steps, or 2.2V to 2.835V in 5mV Steps
- ✓ 500kHz to 1250kHz Configurable Switching Frequency (f_{SW})
- ✓ Differential V_{OUT} Remote Sense

- ✓ Two Low-Dropout (LDO) Regulators:
 - V_{OUT_1V} : 0.9V to 1.2V in 0.1V Steps
 - $V_{OUT_1.8V}$: 1.7V to 1.9V in 0.1V Steps
- ✓ Fully Compliant with I²C and I³C Specifications
- ✓ Accurate V_{OUT} , Output Current (I_{OUT}), and Junction Temperature (T_J) Monitoring
- ✓ Open-Drain Power Good (PG) Indication and General Status Interrupt
- ✓ Configurable Slave Address via the Multiple-Time Programmable (MTP) Memory
- ✓ Selectable Pulse-Frequency Modulation (PFM)/Pulse-Width Modulation (PWM) Mode, Adjustable Frequency, and Current Limit
- ✓ Pre-Biased Start-Up
- ✓ Over-Current Protection (OCP)
- ✓ Under-Voltage Protection (UVP)
- ✓ Under-Voltage Lockout (UVLO)
- ✓ Thermal Shutdown
- ✓ Over-Voltage Protection (OVP)
- ✓ Available in a QFN-35L (5mmx5mm) Package

Show less ^

Active Part Numbers:

MPQ8895FGU-xxxx-Z MPQ8895FGU-xxxx-P

Meaning of P & Z ^

MPQ8895F Resources

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SERVER DDR5

● APPLICATION

DDR5 SDRAM (synchronous dynamic random-access memory) is the latest...

SERVERS (DDR5 POWER)

● APPLICATION

DDR5 SDRAM (synchronous dynamic random-access memory) is the latest...

POWER MANAGEMENT IC (PMIC)

● PRODUCT CATEGORY

TAMING THE BEAST: MEMORY EFFICIENCY IN A...

● ARTICLE

The planet is facing a crisis in energy demand versus supply, and datacenters are at the...

MP5519

● PRODUCT

2.65V to 16V, 6A, Protection IC with

Design Resources

Symbol, Footprint & 3D Model

30+ more formats ^

 Symbols (36)

 Footprint (34)

 3D Models (15)

EDA model is not yet available for this part.
Please enter your email address and we will notify you when it is released.

Email

Submit Request

Buy direct from MPS

Quote or Technical Support

Technical Forum

Questions answered by MPS engineers in 24h

MPM3606 Inductor Value

Hi, I want to estimate the output ripple of MPM3606. The section "Selecting the Output Capacitor" of the datasheet provides some formulas for this. Th...

DC-DC POWER CONVERTERS

Latest activity 12 hours ago 1 Comment

MP5470BGL-0000 default reference voltages and unused outputs

On this PMIC MP5470BGL-0000, I have a few questions. Are the default voltages for this device always factory set at Vref1 = 0.7 V while Vref2=Vref3...

DC-DC POWER CONVERTERS

Latest activity a day ago 4 Comments

MP8030, turning on POE Mosfet

Hi, I'm solving problem with turning on an external poe mosfet. In my design, I used the mosfet ISC320N12LM6ATMA1, which has Qg = 4.1nC @ 0 to 4.5V an...

DC-DC POWER CONVERTERS

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All MPS parts are lead-free, halogen-free and adhere to the RoHS directive. See Environmental section for details.

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